THE ABERDARE TECHNICAL COLLEGE

ICT TECHNICIAN LEVEL 5

CAT 2 TERM 3

87%

UNIT: NETWORK DESIGN AND MANAGEMENT
UNIT CODE: 0612 451 07A
TIME: 1 Hour 30 Minutes
TOTAL MARKS: 50 Marks

NAME: KARU MARCUS

REG NO: ...

DATE: 15/6/2026

**INSTRUCTIONS**

- ✓ *Answer ALL questions.*
- ✓ *Answer in a clear and well-structured manner.*
- ✓ *Show all relevant technical knowledge.*

1. Differentiate between a switch and a router in terms of function and OSI layer operation. (2 Marks)
2. Explain TWO considerations when selecting transmission media for a network installation. (2 Marks)
3. State TWO differences between fibre optic and copper-based network cables. (2 Marks)
4. Describe the role of access points in wireless networking and how they integrate with wired LANs. (2 Marks)
5. Give TWO examples of network security devices and explain their functions in a network. (2 Marks)
6. Differentiate between networking software and network management tools with suitable examples. (2 Marks)
7. Explain TWO functions of a server in a client-server network environment. (2 Marks)
8. State TWO uses of storage systems in a network environment. (2 Marks)
9. Explain the purpose of a multimeter in network installation and troubleshooting. (2 Marks)
10. Differentiate between a patch cord and a backbone network cable in terms of usage. (2 Marks)
11. Describe the function and ONE safety precaution for each of the following tools:
    a) Cable crimpers (2 Marks)
    b) Cable strippers (2 Marks)
    c) Punch down tool (2 Marks)
    d) Network tester (2 Marks)
    e) Tone generator and probe (2 Marks)
12. Explain THREE safety precautions when handling network installation tools and equipment. (3 Marks)
13. State TWO applications of fiber optic installation tools in structured cabling. (2 Marks)



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1. Switch connects devices in LAN using MAC address and it is in layer 2 while router is in layer 3 of OSI model and connects different networks using IP address.

2. Speed range of the media
Cost and environmental condition

3. Fibre optic have higher speed and support data for long distances
Copper cables have limited speed and distance.

4. Connects wireless devices to wired LAN

5. Firewall → Controls traffic in a network.
IDS/IPS → Detects and Prevent intrusion attacks.

6. Networking Software enable communication while networking management tools monitor and control networks

7. Provides shared resources
Manages user authentication and security.

8. Centralized file storage and sharing
Data backup and recovery

9. Used to test voltage continuity and resistance in cables

10. Patch cord are short cable used to connect devices to switch/Patch panels while Backbone cable are high capacity cable connecting different network segm.

11. Cable crimper  Attaches connectors to cables (RJ45)
Wear gloves to prevent cuts from sharp parts.

18

Cable stripper  Remove outer cable insulation without damaging wires.
Use the proper size to avoid being cut.
Punchdown tool  Terminate wires into Patch panels.
Place the blade safely to avoid cutting yourself.
Network tester  Test cable continuity.
Ensure the device is off before connecting.
Tone generator and Probe  Traces and identify cables in walls.
Avoid using near live electrical circuits.

09

12 Wear safety goggles and gloves.
Disconnect Power before working on cables.
Use correct tools only.

03

13 Cutting and stripping fibre cables cleanly.
Testing and splicing fible connection for proper signal transmission.

05

14. Network cables. carry data signals between devices
RJ45 sockets. Provide connection points for plugging in network cables.
Patch cords. Short cables used to connect devices to wall sockets or switches.
Cable trunking covers: Protect and hide cabes along walls.
Cable ties  Neatly arrange and secure cables.

10

15 Network ~~Socket~~. Cables.
Switch.
RJ45 connectors.
Router.

05

16. Compatibility → ensure that all these devices can work together with exiting equipment
Cost : Ensure that the components cost can ~~equivalent~~ equal the Performance.

05

30

17 Plan cable route and install trunking
Pull network cables through trunking
Terminate cables with RJ45 at both ends
Install wall sockets and patch panel
Connect devices to switch using patch cords
Test connectivity

18. Cable damage -> Use proper trunking and pull carefully
Poor connectivity -> Proper termination and test with cable tester

19 Crimping tool -> used to connect RJ45 to cables
Cable tester -> Test cable continuity

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